

BILMA, NigerWMO: **610170**Lat: **18.683N**Lon: **12.917E**Elev: **357**StdP: **97.11**Time Zone: **1.00 (E01)**Period: **86-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.3	6.2	-8.1	2.0	24.3	-7.5	2.1	23.9	15.1	16.5	12.1	16.9	0.0	100	0.435

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	15.6	44.3	21.3	43.5	21.2	42.7	21.1	25.2	37.2	24.5	36.9	24.0	36.7	1.9	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.0	17.4	31.4	20.9	16.3	31.6	20.0	15.3	32.2	78.9	36.1	76.1	36.8	73.8	36.6	30.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.3	9.4	7.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			-0.9	27.2	1.0	2.0	-1.6	28.6	-2.2	29.7	-2.8	30.9	-3.6	32.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.9	16.9	20.9	25.5	30.2	33.6	35.1	34.8	34.2	33.1	28.5	22.8	18.3	(6)
	DBStd	7.00	2.97	3.47	3.47	3.33	2.02	2.20	1.66	1.71	1.75	2.96	2.86	3.03	(7)
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD18.3	118	64	12	1	0	0	0	0	0	0	0	2	39	(9)
	CDD10.0	6516	214	305	479	607	731	753	769	751	693	573	383	257	(10)
	CDD18.3	3592	21	83	222	357	472	503	511	493	443	314	135	37	(11)
	CDH23.3	55187	342	1098	2832	5246	7446	8321	8449	7964	6858	4341	1739	551	(12)
	CDH26.7	36769	92	468	1548	3416	5230	6063	6032	5529	4689	2694	841	167	(13)
(14) Wind	WSAvg	2.0	2.3	2.4	2.7	2.5	1.9	1.7	1.5	1.5	1.4	1.9	1.6	2.3	(14)
(15) Precipitation	PrecAvg	11	0	0	0	1	0	1	2	6	1	0	0	0	(15)
	PrecMax	40	1	0	0	15	12	6	20	31	5	6	0	5	(16)
	PrecMin	0	0	0	0	0	0	0	0	0	0	0	0	0	(17)
	PrecStd	10	0	0	0	3	2	2	4	8	1	1	0	1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.5	36.1	39.7	43.4	45.1	45.7	44.4	43.7	43.2	41.0	36.6	32.4	(19)
		MCWB	15.9	17.1	18.8	20.2	20.9	21.3	22.0	22.7	22.1	20.2	18.0	16.1	(20)
	2%	DB	29.2	34.1	37.9	41.9	44.0	44.7	43.3	42.6	42.1	39.7	35.2	30.3	(21)
		MCWB	13.8	16.0	17.6	19.4	20.3	21.5	21.9	22.4	21.5	19.7	17.4	14.5	(22)
	5%	DB	27.1	31.9	36.2	40.8	43.0	43.7	42.4	41.6	41.3	38.5	33.5	28.6	(23)
		MCWB	12.7	15.0	17.1	18.9	19.9	21.3	21.9	22.6	21.2	19.5	16.7	14.2	(24)
	10%	DB	25.2	29.8	34.1	39.3	41.7	42.7	41.4	40.5	40.4	36.8	31.8	26.6	(25)
		MCWB	12.1	14.1	16.1	18.4	19.6	21.1	21.8	22.7	21.0	18.8	16.0	13.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.3	18.2	20.5	22.5	23.9	24.7	26.1	27.2	24.8	21.9	19.2	16.6	(27)
		MCDB	31.5	34.3	36.2	37.9	38.1	40.3	38.6	37.5	37.9	38.2	34.0	30.9	(28)
	2%	WB	14.3	16.6	18.8	20.9	22.6	23.9	24.7	25.5	24.0	20.6	18.0	15.3	(29)
		MCDB	27.5	32.1	35.2	37.2	37.8	39.5	37.6	36.8	37.9	37.3	33.5	29.1	(30)
	5%	WB	13.1	15.5	17.7	19.9	21.6	23.2	24.0	24.8	23.4	19.9	17.1	14.4	(31)
		MCDB	25.7	30.7	34.5	38.2	38.4	38.8	37.4	36.1	37.1	36.8	32.7	27.6	(32)
	10%	WB	12.3	14.4	16.7	19.0	20.7	22.6	23.4	24.3	22.7	19.2	16.2	13.5	(33)
		MCDB	24.4	29.2	33.1	37.3	39.0	38.4	36.8	35.8	36.2	35.8	31.0	25.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	17.0	17.3	16.9	16.8	16.6	15.6	13.9	12.9	15.5	17.1	18.1	16.6	(35)
		MCDBR	19.3	19.4	18.9	17.8	18.8	16.7	15.2	14.2	16.5	17.3	19.2	18.9	(36)
	5% WB	MCWBR	9.0	8.4	7.8	6.7	6.5	5.9	5.3	4.9	5.8	6.4	8.2	8.7	(37)
		MCWBR	18.3	18.5	18.4	17.0	15.7	13.5	13.0	12.2	13.9	16.5	18.5	17.4	(38)
(40) Clear-Sky Solar Irradiance	taub		0.478	0.530	0.593	0.685	0.728	0.734	0.724	0.736	0.695	0.610	0.492	0.470	(40)
		taud	1.759	1.639	1.514	1.395	1.330	1.344	1.396	1.396	1.423	1.526	1.713	1.755	(41)
	Ebn at Noon		786	770	741	680	644	634	642	637	656	693	765	776	(42)
		Edh at Noon	211	251	294	335	353	345	328	330	318	278	220	206	(43)
(44) All-Sky Solar Radiation	RadAvg		5.38	6.05	6.82	7.19	7.43	7.38	7.24	6.82	6.72	6.21	5.49	5.03	(44)
	RadStd		0.12	0.19	0.33	0.28	0.27	0.24	0.17	0.20	0.16	0.16	0.12	0.14	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page